Linear mixed effects models

Last time

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

- u_i is a *random intercept*
- We use subscripts i and j for $Price_{ij}$, $Satisfaction_{ij}$, and ε_{ij} because they are different for every observation in the data
- We only need a subscript i (neighborhood) for u_i, because there is one random intercept per neighborhood

$\beta_0 + u_i$ = expected price in neighborhood i for a rental w/ 0 satisfaction

Last time: Class activity

β_0 average across all neighborhoods

```
Fixed effects:
                      Estimate Std. Error t value
(Intercept)           27.275     30.547   0.893
overall_satisfaction  14.809      6.303   2.350
```

$\hat{\beta}_0$ $\hat{\beta}_1$

> Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

$\hat{\beta}_0 = 27.28$ on average (across all neighborhoods) we estimate a price of \$27.28 for a rental with 0 satisfaction (not super meaningful since 2.5 is smallest satisfaction in data)

$\hat{\beta}_1 = 14.81$ For a fixed neighborhood, a increase of 1 point in satisfaction is associated with an average increase of \$14.8) in price

Last time: Class activity

```
Random effects:
 Groups       Name        Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
```

> Calculate and interpret the estimated intra-class correlation.

$$\hat{\rho}_{grup} = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \hat{\sigma}_\varepsilon^2}$$

$$= \frac{1048}{1048 + 6762} = 0.134$$

About 13.4% variability in rental price can be explained by differences in price between neighborhoods (after accounting for overall satisfaction)

Warmup

Work on the warmup activity (handout).

Warmup

> What are the groups in the data?

musicians

(multiple rows — performances — for each musician)

id	na	large
1	3	1
1	4	0
1	1	1

Warmup

> What mixed effects model should the researchers use to investigate their hypothesis?

$$\text{Anxiety}_{ij} = \beta_0 + \beta_1 \text{large}_{ij} + u_i + \varepsilon_{ij}$$

anxiety for musician i before performance j

i = musician

j = performance

u_i = variability in anxiety between musicians

ε_{ij} = variability in anxiety within musicians, between different performances

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

> What assumptions are we making in this mixed effects model?

- effect of satisfaction is the same for all neighborhoods (same β_1 for each neighborhood)
- shape: overall relationship is linear (and ↻)
- normality: u & ε are Normal
- constant variance for both u & ε
- neighborhoods are independent
- after accounting for random effect (u_i), independence within neighborhoods

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Shape:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Independence:**
  - random effects are independent
  - observations within neighborhoods are independent after accounting for the random effect (i.e., the random effect captures the correlation within neighborhoods)
  - observations from different neighborhoods are independent

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Normality:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$
- **Constant variance:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

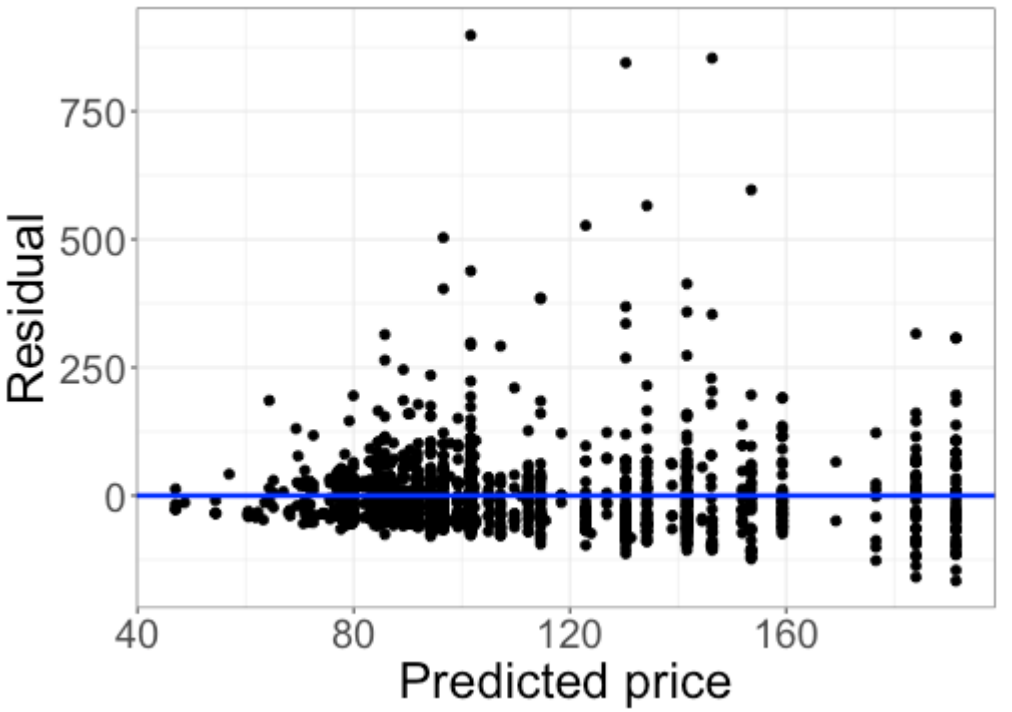
- **Shape assumption:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood
- **Constant variance assumption:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

> How do you think we could check the shape and constant variance assumptions?

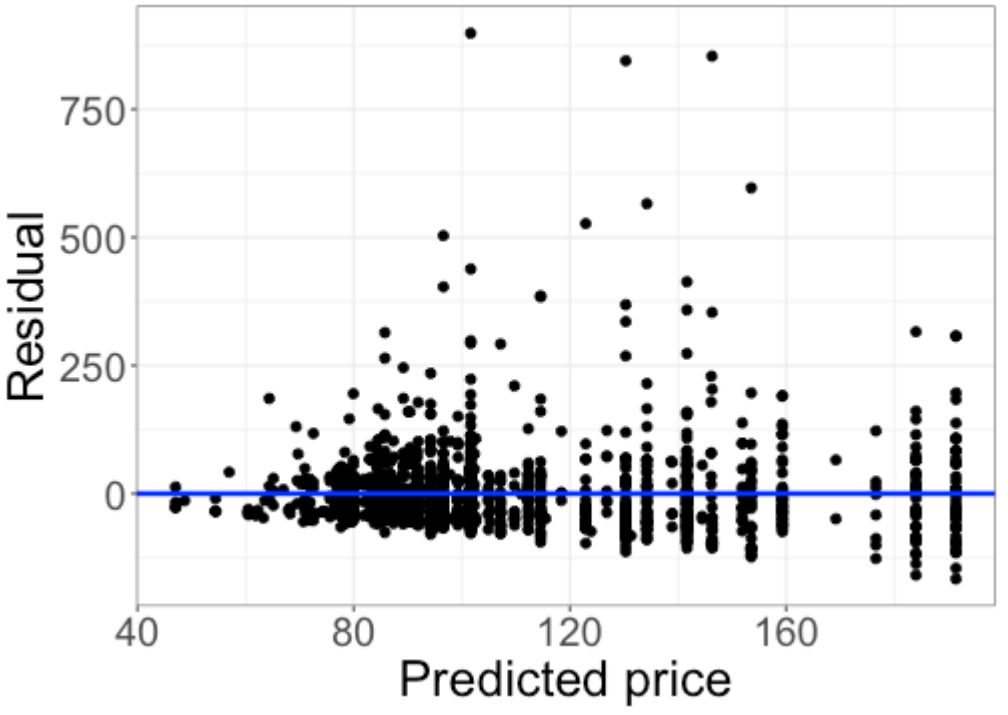
Residual plots

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

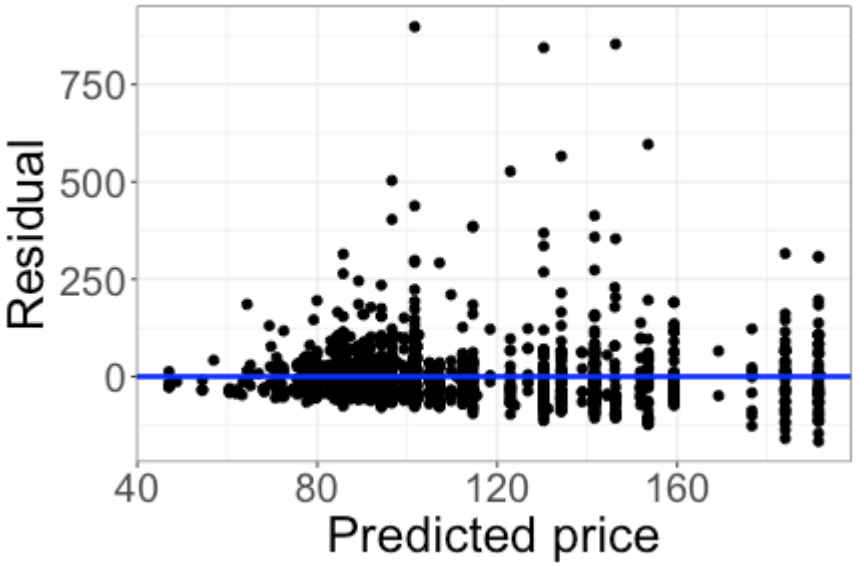


Residual plots



Do the shape and constant variance assumptions look reasonable?

Residual plots



- Linearity seems reasonable
- We can't assess (yet) whether the slope is actually the same for all neighborhoods
- Constant variance may be slightly violated (but probably not enough to matter)

Calculating predicted values

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

> Where do the $\widehat{u}_i$ come from?

Estimated random effects

R calculates an estimated random effect for each group (i.e., neighborhood):

```
m1 <- lmer(price ~ overall_satisfaction +
             (1 | neighborhood),
           data = bnb)
coef(m1)
```

$\hat{\beta}_0 + \hat{u}_i$ $\hat{\beta}_1$

```
## $neighborhood
##                   (Intercept) overall_satisfaction
## Albany Park          16.367331                14.80912
## Archer Heights        9.863461                14.80912
## Avondale             40.533851                14.80912
## Beverly              21.046464                14.80912
## Bridgeport           13.304517                14.80912
## Brighton Park        28.548742                14.80912
## Burnside             12.741349                14.80912
## Calumet Heights      11.465091                14.80912
```

Estimated random effects

this $- \hat{\beta}_0 = \hat{u}_i$

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

> How do I get the random effect estimates $\widehat{u}_i$?

The intercept for each neighborhood is $\widehat{\beta}_0 + \widehat{u}_i$

So subtract the estimate $\widehat{\beta}_0$ from the intercept for each neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Normality assumption:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

> How do you think we could check the normality assumption?

QQ plots

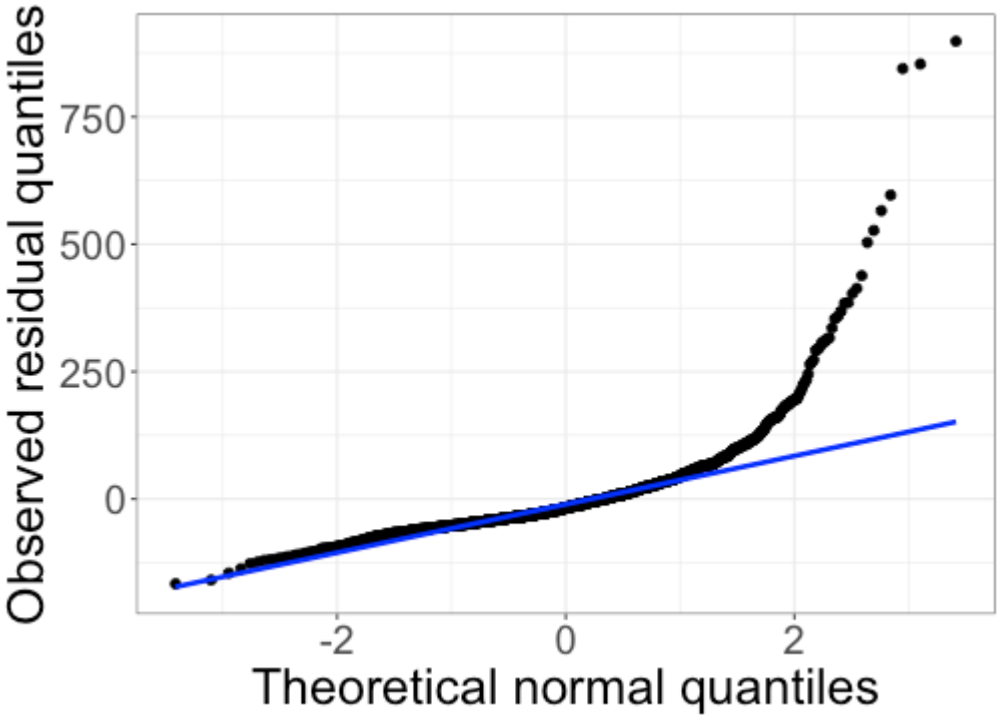
Assumption: $u_i \sim N(0, \sigma_u^2)$

- Check whether the random effect estimates $\widehat{u}_i$ appear normal with a QQ plot

Assumption: $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

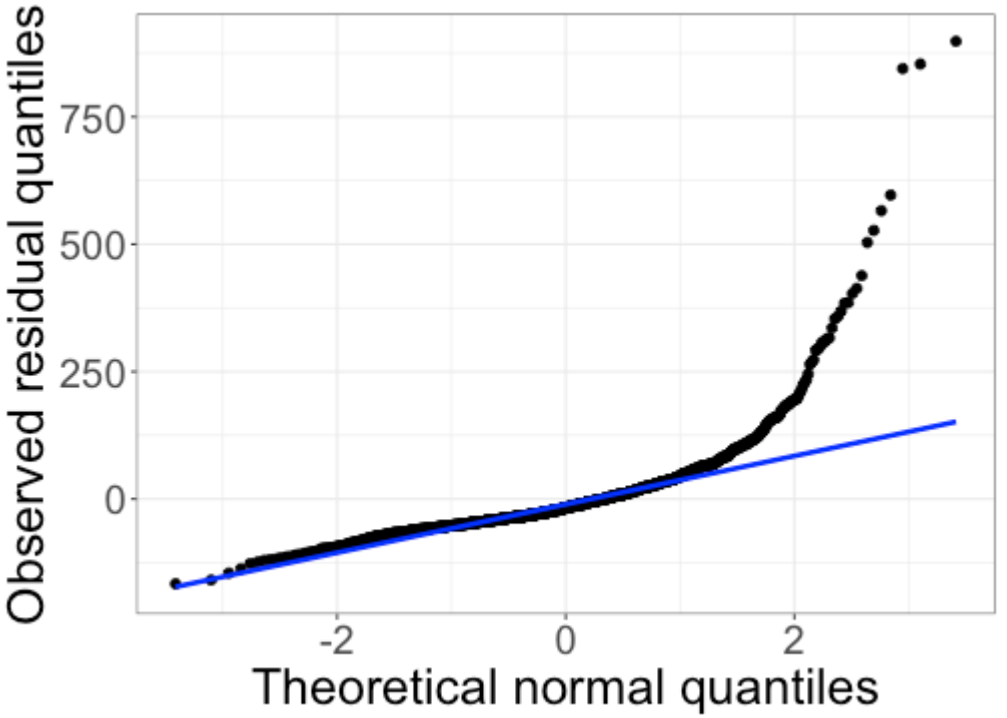
- Check whether the residuals appear normal with a QQ plot

QQ plot for the residuals



Do the residuals appear normal?

QQ plot for the residuals

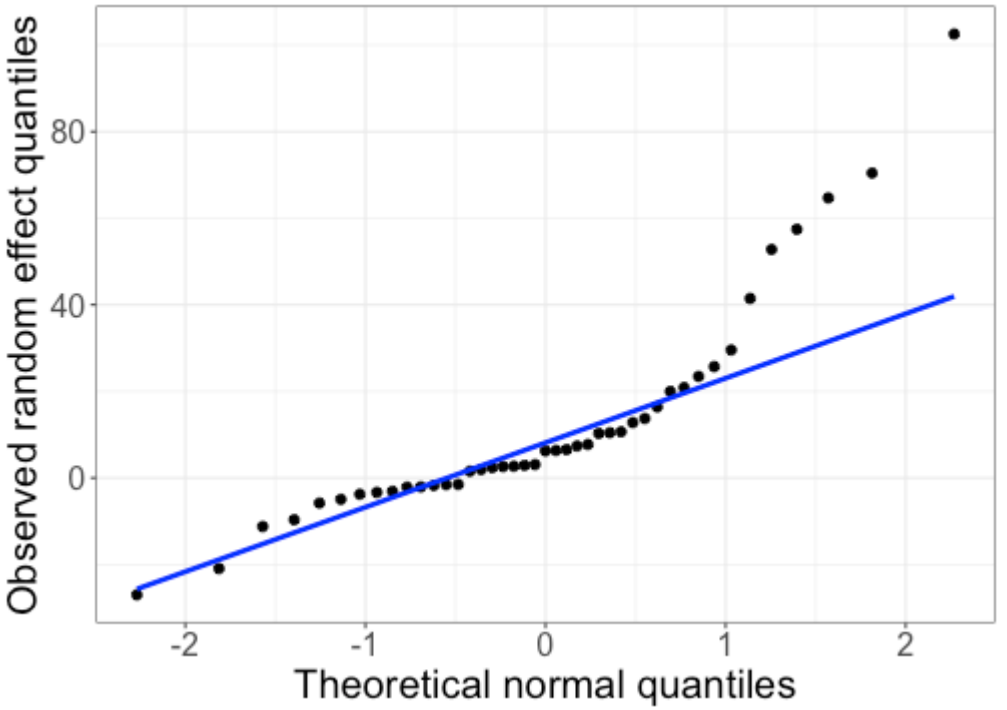


> Do the residuals appear normal?

No -- though the sample size is large, so it might not matter

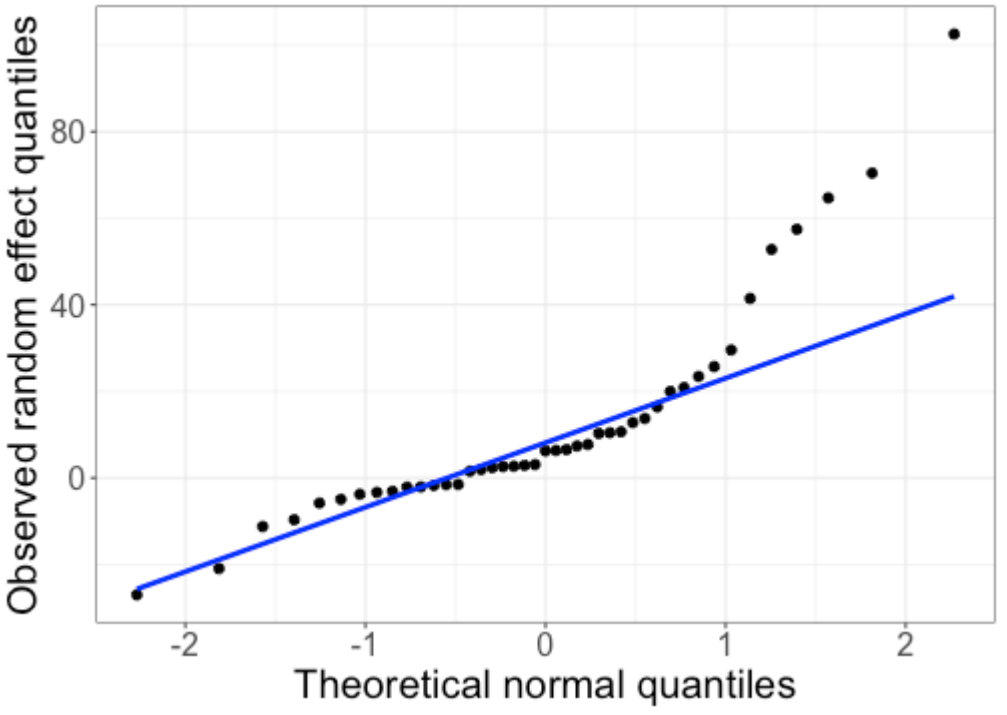
(or try transforming e.g. log(Price))

QQ plot for the random effects



Do the random effects appear normal?

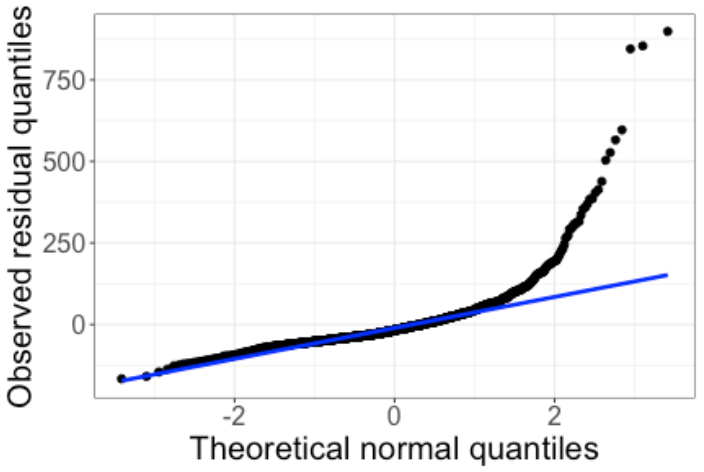
QQ plot for the random effects

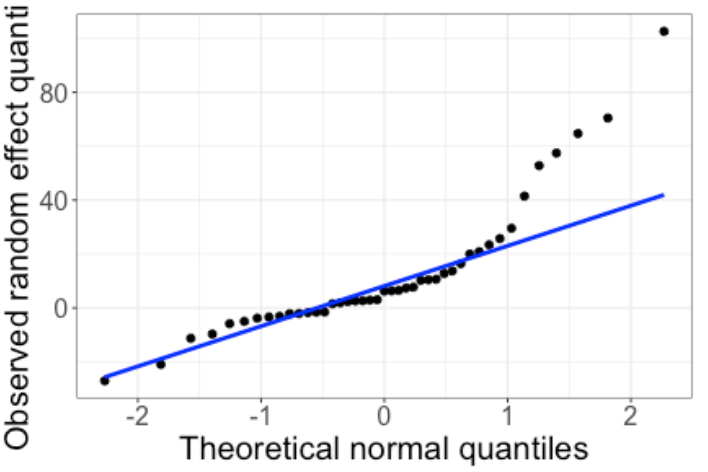


> Do the random effects appear normal?

No -- and this could be more problematic

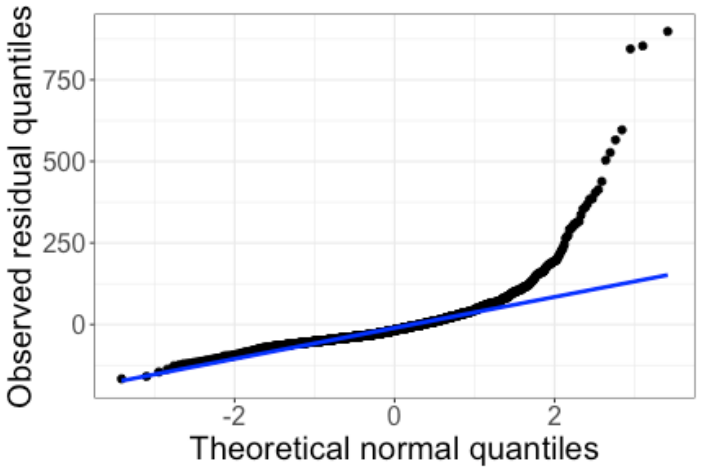
Addressing assumption violations

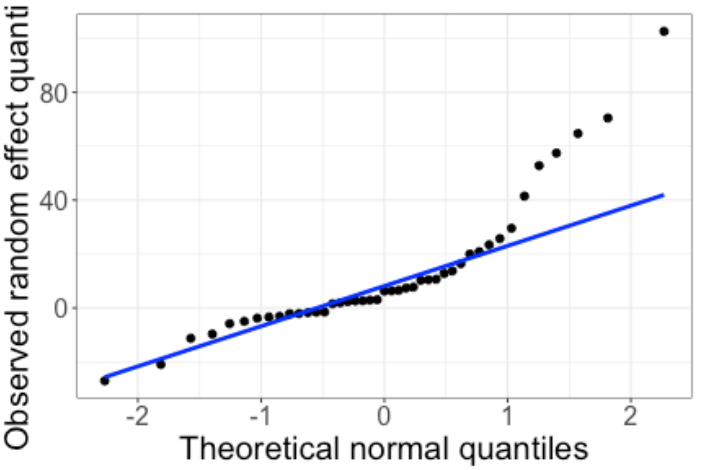




How could we address violations of the normality assumptions?

Addressing assumption violations





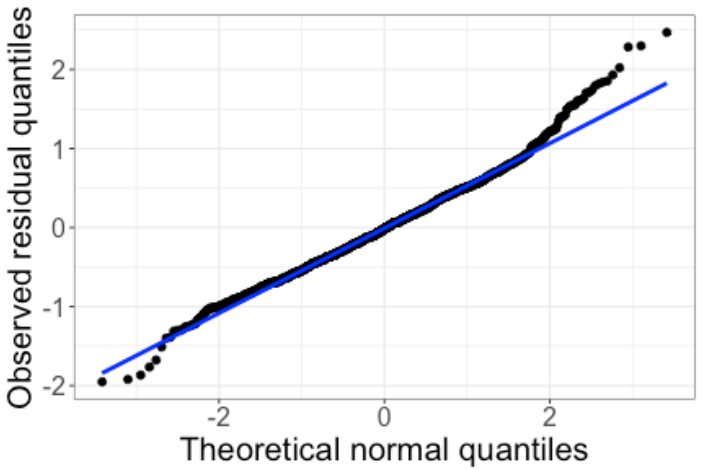
> How could we address violations of the normality assumptions?

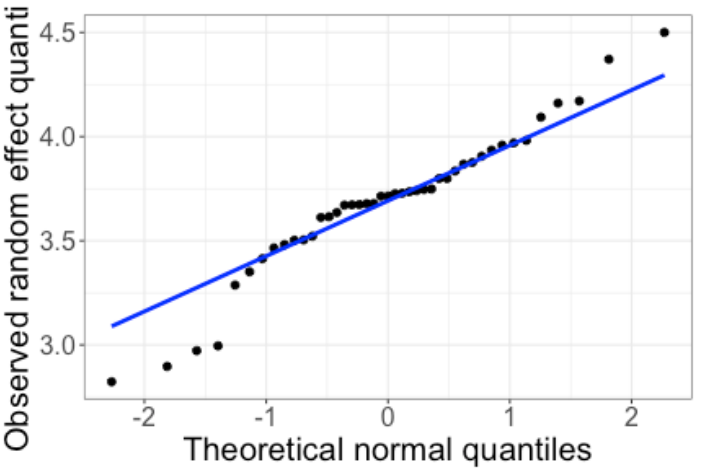
Transformations!

Transformations

$$\log(Price_{ij}) = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$





- Transformations can help with both the residuals and random effects

Class activity

Work on the class activity on the course website. Submit your HTML on Canvas when finished.